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337313 (37)

BE (3rd Semester)

Examination, Nov.-Dec., 2013

Branch : Mechanical

MATERIAL SCIENCE

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Question (a) is compulsory. Solve any two from b, c & d.

Q. 1. (a) Define 'lattice parameter'. 2

(b) A F.C.C. unit cell has a lattice constant a = 4.0×10^{-10} m. Calculate the number of atoms

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per unit area on (110) and (1, 1, 1) planes and density of atoms per unit length in directions [110] and [111]. 7

(c) Explain the effect of grain size on properties of metals. Give one example. 7

(d) Explain the procedure for finding Miller Indices. 7

Q. 2. (a) Define Recrystallisation Temperature. 2

(b) Discuss how the following factors influence the mode of fracture in metals : 7

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(i) Grain size

(ii) Heat treatment

(iii) Low temperature

(c) Distinguish between cold and hot working of metals. 7

(d) How does recovery differ from recrystallisation in a metal ? 7

Q. 3. (a) Define an alloy. 2

(b) Explain peritectic transformation with neat diagram. 7

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(c) What is an equilibrium diagram ? State its importance and objectives. 7

(d) Explain briefly the process of crystallisation of pure metal. 7

Q. 4. (a) State objective of heat treatment. 2

(b) Explain briefly the following annealing processes : 7

(i) Full annealing

(ii) Isothermal annealing

(iii) Sub-critical annealing

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(c) State the effect of tempering temperature on the mechanical properties of steel.

(d) Explain briefly the "pack hardening" of case hardening. 7

Q. 5. (a) Give the composition of grey cast iron and nodular cast iron. 2

(b) State the effects of adding following alloying elements to steel : Nickel, Chromium, Manganese, Vanadium & Silicon. 7

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(c) Give the composition, properties and uses of the following materials :

(i) Bronze

(ii) Bearing metals

(d) Explain the manufacturing of sintered carbide.

Also state its composition and uses.

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